



Installation Instruction

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New Exhaust Gas Temperature Sensor and Injector Quadrant Wiring Harness and Mounting Bracket Installation Instructions

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Product Affected

Engines:

- QSK60 CM850 MCRS
- QSK60 CM2150 MCRS

Applications:

- Industrial (main engine wiring harness with extruded rail design **only**).

Introduction

The purpose of this document is to outline the fitment procedure for the new exhaust gas temperature sensor and injector wiring harness and mounting brackets. For more information, see Technical Service Bulletin, New Exhaust Gas Temperature Sensor and Injector Quadrant Wiring Harness and Mounting Brackets, TSB160095.

Parts Required

The parts required to install are provided in Table 1.

Table 1, Required Parts

Part Description	Part Number	Engine Bank	Quantity
HARNESS,ETR CNT MDL WRG	5373047	Left Bank	2
HARNESS,ETR CNT MDL WRG	5373048	Right Bank	2

Personal Protection Equipment (PPE)

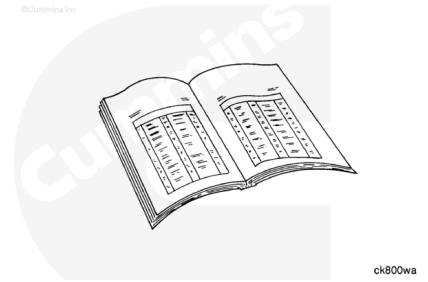
- Completing a Job Safety Assessment (JSA) prior to performing work helps identify job safety hazards and prevent incidents. Use the guidelines below to assess if a situation is safe or at risk prior to performing designated work.
- To reduce the possibility of personal injury, personal protective equipment (PPE) **must** be utilized. Reference General Safety Instructions Procedure 204-006 in Section I of the corresponding Service Manual.

Preparatory Steps

⚠ WARNING ⚠

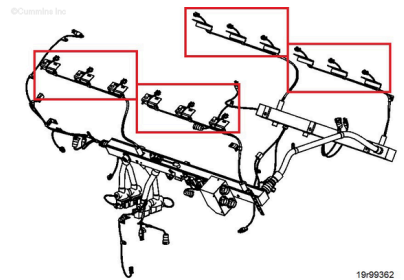
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See original equipment manufacturer (OEM) service information.



Remove

- Remove exhaust gas temperature sensor and injector wiring harnesses.
 - See QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/05/05-019-043.html) in Section 19.
 - See QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) in Section 19.
- Remove capscrews that secure exhaust gas temperature sensor and injector wiring harnesses to cylinder heads.
- Remove exhaust gas temperature sensor and injector wiring harness and mounting bracket assembly.
- Each exhaust gas temperature sensor and injector wiring harness quadrant has two mounting brackets and four capscrews.

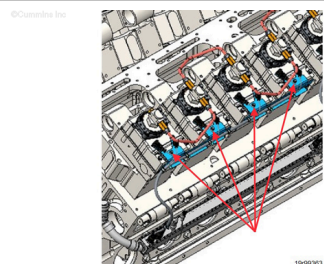


Install

- Install exhaust gas temperature sensor and injector wiring harness and mounting bracket assembly and capscrews.

Torque Value: 18 n•m [160 in-lb]

- Connect exhaust gas temperature sensor and injector connectors to exhaust gas temperature sensor and injector wiring harness connectors.



Left and Right Engine Bank Quadrants (Front) **Only**

- Route main exhaust gas temperature sensor and injector wiring harness connections between cylinder four and cylinder five.
- Use low temperature aftercooler (LTA) coolant outlet flange capscrew to mount p-clip.

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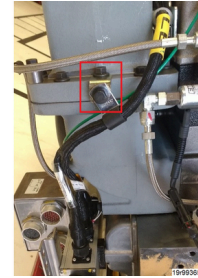


Torque Value: 45 n·m [33 ft-lb]

Left and Right Engine Bank Quadrants (Rear) **Only**

- Route main exhaust gas temperature sensor and injector wiring harness connections over cylinder eight.
- Use aftercooler housing capscrew to mount p-clip adaptor.

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Aluminum Aftercooler Cover Assembly

Torque Value: 45 n·m [33 ft-lb]

Cast Iron Aftercooler Cover Assembly

Torque Value: 65 n·m [48 ft-lb]

- Fit p-clip to mounting adaptor.

Torque Value: 45 n·m [33 ft-lb]

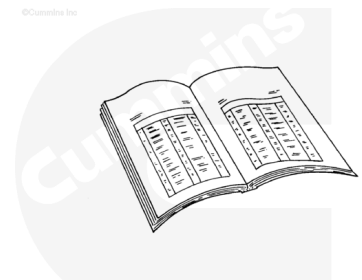
- Connect exhaust gas temperature sensor and injector wiring harness connector to main engine wiring harness connection.

Finishing Steps

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⚠ WARNING ⚠

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- Connect batteries. See OEM service information.

Associated Publications

See Technical Service Bulletin, New Exhaust Gas Temperature Sensor and Injector Quadrant Wiring Harness and Mounting Brackets, TSB160095.

Table 2, Associated Procedures					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Troubleshooting and Repair Manual	QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System)	4021533	Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsyss2/xml/en/procedures/05/05-019-043.html)	19
Electronic Control System Troubleshooting and Repair Manual	QSK38, QSK50, and QSK60 CM2150	4022102	Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsyss2/xml/en/procedures/122/122-019-043.html)	19

Document History

Date	Details
2016-10-7	Module Created

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